$C_{e\phi_2}^{(22)}$ i1_i2_ →

$$
\begin{aligned}
&\frac{1}{16\pi^2}\Big(-\frac{1}{9} g_2^4 y_e^{i1i2} \mathrm{LF}_{3,0}[m_2] - \frac{1}{6} g_2^4 y_e^{i1i2} \mathrm{LF}_{4,-1}[m_2] + \frac{8}{45} g_2^4 y_e^{i1i2} \mathrm{LF}_{5,-2}[m_2] + \frac{1}{72} g_2^2 y_e^{i1i2} \\
&\left(-36 \overline{y_e}^{pr} y_e^{pr} + 5 \textstyle\sum_p g_2^2\right) \mathrm{LF}_{3,0}[m_{\tilde{l}}^{p}] - \frac{1}{48} g_2^2 y_e^{i1i2} \left(-24 \overline{y_e}^{pr} y_e^{pr} + \textstyle\sum_p g_2^2\right) \mathrm{LF}_{4,-1}[m_{\tilde{l}}^{p}] - \\
&\frac{2}{45} \textstyle\sum_p g_2^4 y_e^{i1i2} \mathrm{LF}_{5,-2}[m_{\tilde{l}}^{p}] + \frac{1}{24} g_2^2 y_e^{i1i2} \left(-36 \overline{y_d}^{pr} y_d^{pr} + 5 \textstyle\sum_p g_2^2\right) \mathrm{LF}_{3,0}[m_{\tilde{q}}^{p}] - \\
&\frac{1}{16} g_2^2 y_e^{i1i2} \left(-24 \overline{y_d}^{pr} y_d^{pr} + \textstyle\sum_p g_2^2\right) \mathrm{LF}_{4,-1}[m_{\tilde{q}}^{p}] - \frac{2}{15} \textstyle\sum_p g_2^4 y_e^{i1i2} \mathrm{LF}_{5,-2}[m_{\tilde{q}}^{p}] - \\
&\frac{1}{18} g_2^4 y_e^{i1i2} \mathrm{LF}_{3,0}[\tilde{\mu}] - \frac{1}{12} g_2^4 y_e^{i1i2} \mathrm{LF}_{4,-1}[\tilde{\mu}] + \frac{4}{45} g_2^4 y_e^{i1i2} \mathrm{LF}_{5,-2}[\tilde{\mu}] + \\
&g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{e}}^{r}] - 2 g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{e}}^{r}] + \\
&g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{e}}^{r}] - \frac{1}{4} g_1^4 y_e^{i1i2} \mathrm{LF}_{2,2,-1}[m_1, m_{\tilde{e}}^{i2}] + \\
&\frac{1}{4} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{l}}^{p}] - \frac{1}{2} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{l}}^{p}] + \\
&\frac{1}{4} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{l}}^{p}] + \frac{1}{32} g_1^2 y_e^{i1i2} \left(g_1^2 - g_2^2\right) \mathrm{LF}_{2,2,-1}[m_1, m_{\tilde{l}}^{i1}] + \\
&\frac{3}{4} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,0}[m_2, m_{\tilde{l}}^{p}] - \frac{3}{2} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,-1}[m_2, m_{\tilde{l}}^{p}] + \\
&\frac{3}{4} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{4,1,-2}[m_2, m_{\tilde{l}}^{p}] + \frac{1}{32} g_2^2 y_e^{i1i2} \left(3 g_1^2 + g_2^2\right) \mathrm{LF}_{2,2,-1}[m_2, m_{\tilde{l}}^{i1}] + \\
&\frac{2}{3} g_2^4 y_e^{i1i2} \mathrm{LF}_{2,1,0}[m_2, \tilde{\mu}] + \frac{17}{12} g_2^4 y_e^{i1i2} \mathrm{LF}_{2,2,-1}[m_2, \tilde{\mu}] - \frac{1}{3} g_2^4 y_e^{i1i2} \mathrm{LF}_{3,1,-1}[m_2, \tilde{\mu}] - \\
&\frac{7}{3} g_2^4 y_e^{i1i2} \mathrm{LF}_{3,2,-2}[m_2, \tilde{\mu}] + \frac{3}{2} g_2^4 m_2^2 y_e^{i1i2} \mathrm{LF}_{3,2,-1}[m_2, \tilde{\mu}] + g_2^4 y_e^{i1i2} \mathrm{LF}_{3,3,-3}[m_2, \tilde{\mu}] - \\
&g_2^4 m_2^2 y_e^{i1i2} \mathrm{LF}_{3,3,-2}[m_2, \tilde{\mu}] + g_2^4 y_e^{i1i2} \mathrm{LF}_{4,2,-3}[m_2, \tilde{\mu}] - g_2^4 m_2^2 y_e^{i1i2} \mathrm{LF}_{4,2,-2}[m_2, \tilde{\mu}] - \\
&\frac{1}{4} g_1^2 y_e^{i1i2} \overline{a_d}^{pr} a_d^{pr} \mathrm{LF}_{2,2,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] - \frac{1}{4} g_1^2 y_e^{i1i2} \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{2,2,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{4} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,0}[m_{\tilde{e}}^{r}, \tilde{\mu}] - \frac{1}{8} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,2,-1}[m_{\tilde{e}}^{r}, \tilde{\mu}] + \\
&\frac{1}{2} g_1^4 y_e^{i1i2} \mathrm{LF}_{2,1,0}[m_{\tilde{e}}^{i2}, m_1] - \frac{1}{2} g_2^2 y_e^{i1i2} \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \\
&\frac{1}{4} g_1^2 y_e^{i1i2} \overline{a_e}^{pr} a_e^{pr} \mathrm{LF}_{3,2,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \frac{1}{8} y_e^{i1i2} \overline{a_e}^{pr} a_e^{pr} \left(3 g_1^2 + 5 g_2^2\right) \mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] - \\
&\frac{1}{6} y_e^{i1i2} \overline{a_e}^{pr} a_e^{pr} \left(3 g_1^2 + g_2^2\right) \mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \\
&\frac{1}{2} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} \mathrm{LF}_{2,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] - \frac{1}{2} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} \mathrm{LF}_{3,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] - \\
&\frac{1}{4} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,0}[m_{\tilde{l}}^{p}, \tilde{\mu}] + \frac{1}{8} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,2,-1}[m_{\tilde{l}}^{p}, \tilde{\mu}] + \\
&\frac{1}{16} g_1^2 y_e^{i1i2} \left(-g_1^2 + g_2^2\right) \mathrm{LF}_{2,1,0}[m_{\tilde{l}}^{i1}, m_1] - \frac{1}{16} g_2^2 y_e^{i1i2} \left(3 g_1^2 + g_2^2\right) \mathrm{LF}_{2,1,0}[m_{\tilde{l}}^{i1}, m_2] - \\
&\frac{1}{2} y_e^{i1i2} \overline{a_d}^{pr} a_d^{pr} \left(2 g_1^2 + 3 g_2^2\right) \mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] + \frac{1}{4} g_1^2 y_e^{i1i2} \overline{a_d}^{pr} a_d^{pr} \mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] + \\
&\frac{3}{8} y_e^{i1i2} \overline{a_d}^{pr} a_d^{pr} \left(7 g_1^2 + 5 g_2^2\right) \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] - \\
&\frac{1}{2} y_e^{i1i2} \overline{a_d}^{pr} a_d^{pr} \left(3 g_1^2 + g_2^2\right) \mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] + \\
&\frac{3}{2} \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_e^{i1i2} \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] - \frac{3}{2} \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_e^{i1i2} \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] - \\
&\frac{1}{8} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} y_u^{pr} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{2,2,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \frac{3}{2} g_2^2 \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} y_u^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] - \\
&\frac{1}{2} g_2^2 \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} y_u^{pr} \mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] - \frac{3}{2} g_2^2 \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} y_u^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \\
&\frac{1}{4} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} y_u^{pr} \left(g_1^2 - g_2^2\right) \mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \frac{1}{8} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} y_u^{pr} \left(g_1^2 + g_2^2\right) \\
&\quad \mathrm{LF}_{3,2,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \frac{3}{4} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} y_u^{pr} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] - \\
&\frac{1}{2} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} y_u^{pr} \left(3 g_1^2 + g_2^2\right) \mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \frac{1}{2} g_1^2 y_e^{i1i2} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_1] - \\
&\frac{3}{8} g_1^4 y_e^{i1i2} \mathrm{LF}_{3,2,-2}[\tilde{\mu}, m_1] - \frac{3}{8} g_1^4 m_1^2 y_e^{i1i2} \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_1] - \\
&\frac{1}{12} g_1^2 y_e^{i1i2} \left(12 g_1^2 + 7 g_2^2\right) \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_1] + \frac{1}{4} g_1^4 y_e^{i1i2} \mathrm{LF}_{4,2,-3}[\tilde{\mu}, m_1] + \\
&\frac{1}{4} g_1^4 m_1^2 y_e^{i1i2} \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{6} g_1^2 y_e^{i1i2} \left(3 g_1^2 + g_2^2\right) \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_1] + \\
&\frac{1}{6} g_2^2 y_e^{i1i2} \left(9 g_1^2 + g_2^2\right) \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_2] - \frac{85}{24} g_2^4 y_e^{i1i2} \mathrm{LF}_{3,2,-2}[\tilde{\mu}, m_2] - \\
&\frac{3}{8} g_2^4 m_2^2 y_e^{i1i2} \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_2] - \frac{3}{4} g_2^2 y_e^{i1i2} \left(4 g_1^2 + g_2^2\right) \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_2] + \\
&\frac{7}{4} g_2^4 y_e^{i1i2} \mathrm{LF}_{4,2,-3}[\tilde{\mu}, m_2] + \frac{1}{4} g_2^4 m_2^2 y_e^{i1i2} \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \\
&\frac{1}{2} g_2^2 y_e^{i1i2} \left(3 g_1^2 + g_2^2\right) \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_2] + \frac{1}{2} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{2,1,0}[\tilde{\mu}, m_{\tilde{e}}^{r}] - \\
&\overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_{\tilde{e}}^{r}] + \frac{1}{2} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_{\tilde{e}}^{r}] + \\
&\overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{2,1,0}[\tilde{\mu}, m_{\tilde{l}}^{s}] - 2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_{\tilde{l}}^{s}] + \\
&\overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_{\tilde{l}}^{s}] - \frac{1}{2} m_1 g_1^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{2} m_1 g_1^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] - \frac{1}{4} m_1 g_1^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \\
&\quad \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] - \frac{1}{8} m_1 g_1^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] + \\
&\frac{1}{4} m_1 g_1^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] + g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{e}}^{r}, \tilde{\mu}] - \\
&2 g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{e}}^{r}, \tilde{\mu}] + g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}[m_1, m_{\tilde{e}}^{r}, \tilde{\mu}] - \\
&g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}] + \frac{1}{2} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}] - \\
&\frac{1}{4} m_1 g_1^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}] + \frac{1}{8} m_1 g_1^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} \\
&\quad \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}] + \frac{1}{4} m_1 g_1^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{16} m_1 g_1^2 a_e^{i1i2} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}] - \frac{3}{4} g_1^4 y_e^{i1i2} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] - \\
&\frac{1}{2} g_1^4 m_1^2 y_e^{i1i2} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] + \frac{1}{2} g_1^4 y_e^{i1i2} \mathrm{LF}_{3,1,1,-2}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] + \\
&\frac{1}{2} g_1^4 m_1^2 y_e^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] - \frac{1}{8} m_1 g_1^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{i2}] - \\
&\frac{1}{2} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{l}}^{p}, \tilde{\mu}] + g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{l}}^{p}, \tilde{\mu}] - \\
&\frac{1}{2} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}[m_1, m_{\tilde{l}}^{p}, \tilde{\mu}] + \frac{1}{8} m_1 g_1^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{e}}^{r}] + \\
&\frac{1}{16} m_1 g_1^2 a_e^{i1i2} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{e}}^{i2}] - \\
&\frac{1}{4} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{8} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] - \frac{3}{16} g_1^4 y_e^{i1i2} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&\frac{1}{8} g_1^4 m_1^2 y_e^{i1i2} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \frac{1}{8} g_1^4 y_e^{i1i2} \mathrm{LF}_{3,1,1,-2}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \\
&\frac{1}{8} g_1^4 m_1^2 y_e^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \frac{7}{8} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{e}}^{r}] + \\
&\frac{1}{8} g_1^2 \left(2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} + y_e^{i1i2} \left(5 g_1^2 - g_2^2\right)\right) \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] + \\
&\frac{3}{4} g_1^4 m_1^2 y_e^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] - \frac{1}{4} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{16} g_1^2 \left(-2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} + y_e^{i1i2} \left(-2 g_1^2 + g_2^2\right)\right) \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] - \\
&\frac{3}{16} g_1^4 m_1^2 y_e^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \frac{3}{2} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_2, m_{\tilde{l}}^{p}, \tilde{\mu}] - \\
&3 g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_2, m_{\tilde{l}}^{p}, \tilde{\mu}] + \frac{3}{2} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}[m_2, m_{\tilde{l}}^{p}, \tilde{\mu}] - \\
&\frac{1}{4} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_2, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] + \frac{1}{8} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] - \\
&\frac{15}{16} g_2^4 y_e^{i1i2} \mathrm{LF}_{2,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{8} g_2^4 m_2^2 y_e^{i1i2} \mathrm{LF}_{2,1,1,0}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \\
&\frac{5}{8} g_2^4 y_e^{i1i2} \mathrm{LF}_{3,1,1,-2}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \frac{1}{8} g_2^4 m_2^2 y_e^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&2 g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] - \frac{3}{2} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
&g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] + m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \\
&\frac{1}{8} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{e}}^{r}] + \frac{3}{2} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{16} g_2^2 \left(6 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} + y_e^{i1i2} \left(-3 g_1^2 + 22 g_2^2\right)\right) \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \\
&\frac{5}{16} g_2^4 m_2^2 y_e^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \frac{3}{4} \overline{y_d}^{st} y_d^{sr} y_e^{i1i2} \overline{a_d}^{pr} a_d^{pt} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}] - \\
&\frac{3}{4} \overline{y_d}^{st} y_d^{sr} y_e^{i1i2} \overline{a_d}^{pr} a_d^{pt} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] - \frac{1}{2} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \\
&\quad \mathrm{LF}_{2,1,1,-1}[m_{\tilde{e}}^{r}, m_1, m_{\tilde{e}}^{i2}] - \frac{1}{2} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, \tilde{\mu}] - \\
&\frac{1}{4} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, \tilde{\mu}] + \frac{1}{4} \overline{y_e}^{st} y_e^{sr} y_e^{i1i2} \overline{a_e}^{pr} a_e^{pt} \\
&\quad \mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}] + \frac{1}{4} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, \tilde{\mu}] - \\
&\frac{1}{4} \overline{y_e}^{st} y_e^{sr} y_e^{i1i2} \overline{a_e}^{pr} a_e^{pt} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \frac{1}{2} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \\
&\quad \mathrm{LF}_{2,1,1,-1}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{e}}^{r}] - \frac{1}{8} m_1 g_1^2 a_e^{i1i2} \left(5 g_1^2 + g_2^2\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] + \\
&\frac{1}{8} m_1 g_1^2 a_e^{i1i2} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] - \frac{1}{2} g_1^4 y_e^{i1i2} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{e}}^{i2}, m_1, \tilde{\mu}] - \\
&\frac{1}{8} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{l}}^{p}, m_1, m_{\tilde{l}}^{i1}] - \frac{1}{8} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{l}}^{p}, m_2, m_{\tilde{l}}^{i1}] + \\
&\frac{1}{2} \overline{y_e}^{st} y_e^{sr} y_e^{i1i2} \overline{a_e}^{pr} a_e^{pt} \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] - \\
&\frac{1}{2} \overline{y_e}^{st} y_e^{sr} y_e^{i1i2} \overline{a_e}^{pr} a_e^{pt} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] + \\
&\overline{y_e}^{st} y_e^{pt} y_e^{i1i2} \overline{a_e}^{pr} a_e^{sr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] - \overline{y_e}^{st} y_e^{pt} y_e^{i1i2} \overline{a_e}^{pr} a_e^{sr} \\
&\quad \mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] - \frac{1}{2} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{si2} y_e^{i1r} \mathrm{LF}_{1,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, \tilde{\mu}] + \\
&\frac{1}{4} \overline{y_e}^{pr} \overline{y_e}^{st} \left(-y_e^{pt} y_e^{si2} y_e^{i1r} + y_e^{pi2} y_e^{sr} y_e^{i1t}\right) \mathrm{LF}_{2,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, \tilde{\mu}] + \\
&\frac{1}{8} m_1 g_1^2 a_e^{i1i2} \left(g_1^2 - 3 g_2^2\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}] + \\
&\frac{1}{8} m_1 g_1^2 a_e^{i1i2} \left(g_1^2 + g_2^2\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}] + \frac{1}{8} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \\
&\quad \mathrm{LF}_{2,1,1,-1}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{l}}^{p}] + \frac{1}{8} g_1^2 y_e^{i1i2} \left(-g_1^2 + g_2^2\right) \mathrm{LF}_{2,1,1,-1}[m_{\tilde{l}}^{i1}, m_1, \tilde{\mu}] + \\
&\frac{1}{8} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[m_{\tilde{l}}^{i1}, m_2, m_{\tilde{l}}^{p}] + \frac{1}{8} g_2^2 y_e^{i1i2} \left(3 g_1^2 + g_2^2\right) \\
&\quad \mathrm{LF}_{2,1,1,-1}[m_{\tilde{l}}^{i1}, m_2, \tilde{\mu}] + \frac{3}{2} \overline{y_d}^{st} y_d^{sr} y_e^{i1i2} \overline{a_d}^{pr} a_d^{pt} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}] - \\
&\frac{3}{2} \overline{y_d}^{st} y_d^{sr} y_e^{i1i2} \overline{a_d}^{pr} a_d^{pt} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}] + 3 \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \overline{a_d}^{pr} a_d^{sr} \\
&\quad \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] - 3 \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \overline{a_d}^{pr} a_d^{sr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \\
&\quad \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] + \frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \\
&\quad \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] + \frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}] + \\
&\frac{3}{4} g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_2] - \frac{7}{4} g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_2] - \\
&\frac{3}{4} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] + g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{4,1,1,-3}[\tilde{\mu}, m_1, m_2] + \\
&\frac{1}{2} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - \frac{13}{8} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}] + \\
&\frac{3}{4} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}] - \frac{3}{4} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{e}}^{i2}] + \\
&\frac{1}{2} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_{\tilde{e}}^{i2}] + \frac{1}{4} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{l}}^{p}] + \\
&\frac{3}{8} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{l}}^{i1}] - \frac{1}{4} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_{\tilde{l}}^{i1}] + \\
&g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{3,2,1,-3}[\tilde{\mu}, m_2, m_1] + m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] - \\
&\frac{3}{8} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}] + \frac{1}{4} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}] - \\
&3 g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{l}}^{p}] + \frac{3}{2} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_2, m_{\tilde{l}}^{p}] - \\
&\frac{9}{8} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{l}}^{i1}] + \frac{3}{4} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_2, m_{\tilde{l}}^{i1}] + \\
&\frac{1}{4} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pi2} y_e^{sr} y_e^{i1t} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] + \\
&\frac{1}{4} \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{si2} y_e^{i1r} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] + \\
&\frac{3}{8} g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{1,1,1,1,-1}[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \frac{1}{4} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \\
&\quad \mathrm{LF}_{1,1,1,1,0}[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{8} g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{2,1,1,1,-2}[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&\frac{1}{8} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \frac{1}{8} m_1 g_1^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \\
&\quad \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] - \frac{3}{16} m_1 g_1^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] - \\
&\frac{1}{4} g_1^2 y_e^{pi2} \overline{a_e}^{pr} a_e^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] + \frac{1}{8} g_1^2 y_e^{pi2} \overline{a_e}^{pr} a_e^{i1r} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] - \frac{3}{8} m_1 g_1^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \\
&\frac{1}{16} m_1 g_1^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, \tilde{\mu}] - g_1^2 y_e^{i1r} \overline{a_e}^{pr} a_e^{pi2} \\
&\quad \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] + \frac{1}{2} g_1^2 y_e^{i1r} \overline{a_e}^{pr} a_e^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] + \\
&m_1 g_1^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] + g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \\
&\quad \mathrm{LF}_{1,1,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, \tilde{\mu}] + \frac{3}{4} m_1 g_1^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}, \tilde{\mu}] - \\
&\frac{1}{8} m_1 g_1^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}, \tilde{\mu}] + m_1 g_1^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} \\
&\quad \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] + m_1 g_1^4 a_e^{i1i2} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&\frac{1}{2} g_1^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,1,-1}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}] - \frac{1}{8} g_1^2 g_2^2 y_e^{i1i2} \\
&\quad \mathrm{LF}_{2,1,1,1,-2}[m_2, m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{8} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \\
&\frac{1}{8} m_2 g_2^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] - \frac{3}{16} m_2 g_2^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] - \frac{1}{4} g_2^2 y_e^{pi2} \overline{a_e}^{pr} a_e^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{8} g_2^2 y_e^{pi2} \overline{a_e}^{pr} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] + \frac{3}{8} m_2 g_2^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \\
&\quad \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{16} m_2 g_2^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \\
&\frac{1}{2} g_2^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \mathrm{LF}_{1,1,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}] + \frac{3}{4} y_e^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} a_d^{sr} \\
&\quad \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] - \frac{3}{4} y_e^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} a_d^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] - \\
&\frac{1}{2} g_1^2 y_e^{i1r} \overline{a_e}^{pr} a_e^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}^{r}, m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}] + \frac{3}{16} m_1 g_1^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}^{r}, m_1, m_{\tilde{l}}^{p}, \tilde{\mu}] - \frac{1}{16} m_1 g_1^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}^{r}, m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&\frac{1}{16} m_2 g_2^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}^{r}, m_2, m_{\tilde{l}}^{p}, \tilde{\mu}] + \frac{1}{16} m_2 g_2^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}^{r}, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{2} \overline{y_e}^{st} y_e^{si2} y_e^{i1r} \overline{a_e}^{pr} a_e^{pt} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, \tilde{\mu}] + \\
&\frac{1}{4} \overline{y_e}^{st} y_e^{si2} y_e^{i1r} \overline{a_e}^{pr} a_e^{pt} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, \tilde{\mu}] + \\
&\frac{1}{4} y_e^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} a_e^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}] - \\
&\frac{1}{2} \overline{y_e}^{st} y_e^{pi2} y_e^{i1t} \overline{a_e}^{pr} a_e^{sr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, \tilde{\mu}] - \\
&\frac{1}{4} y_e^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} a_e^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] - \\
&\frac{1}{4} \overline{y_e}^{st} y_e^{si2} y_e^{i1r} \overline{a_e}^{pr} a_e^{pt} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] + \\
&\frac{1}{2} g_1^2 y_e^{i1r} \overline{a_e}^{pr} a_e^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] - \\
&\frac{1}{8} m_1 g_1^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{p}, \tilde{\mu}] - \\
&\frac{1}{8} g_1^2 y_e^{pi2} \overline{a_e}^{pr} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}^{p}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] - \\
&\frac{3}{16} m_1 g_1^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}^{p}, m_1, m_{\tilde{e}}^{r}, \tilde{\mu}] + \\
&\frac{1}{8} m_1 g_1^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}^{p}, m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] - \\
&\frac{1}{8} g_2^2 y_e^{pi2} \overline{a_e}^{pr} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}^{p}, m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] + \\
&\frac{1}{16} m_2 g_2^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}^{p}, m_2, m_{\tilde{e}}^{r}, \tilde{\mu}] + \\
&\frac{1}{2} y_e^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} a_e^{sr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}] - \\
&\frac{1}{2} y_e^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} a_e^{sr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}] + \\
&\frac{1}{4} \overline{y_e}^{st} y_e^{pi2} y_e^{i1t} \overline{a_e}^{pr} a_e^{sr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}, \tilde{\mu}] - \\
&\frac{1}{4} \overline{y_e}^{st} y_e^{pi2} y_e^{i1t} \overline{a_e}^{pr} a_e^{sr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] + \\
&\frac{1}{8} g_1^2 y_e^{pi2} \overline{a_e}^{pr} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{16} m_1 g_1^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{r}, \tilde{\mu}] + \\
&\frac{1}{8} g_2^2 y_e^{pi2} \overline{a_e}^{pr} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}^{i1}, m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] - \\
&\frac{1}{16} m_2 g_2^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde{l}}^{i1}, m_2, m_{\tilde{e}}^{r}, \tilde{\mu}] + \\
&\frac{3}{2} y_e^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} a_d^{sr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{2} y_e^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} a_d^{sr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{2} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} a_d^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] + \\
&\frac{3}{2} \tilde{\mu}^4 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] - \\
&\frac{3}{2} \tilde{\mu}^4 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] - \\
&\frac{3}{2} \tilde{\mu}^4 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \\
&\frac{3}{2} \tilde{\mu}^4 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] - \\
&\frac{3}{2} \tilde{\mu}^4 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] - \\
&\frac{3}{4} \tilde{\mu}^4 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] + \\
&\frac{1}{2} g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{2,1,1,1,-2}[\tilde{\mu}, m_1, m_2, m_{\tilde{e}}^{i2}] + \frac{1}{2} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{e}}^{i2}] - \frac{1}{8} g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{2,1,1,1,-2}[\tilde{\mu}, m_1, m_2, m_{\tilde{l}}^{i1}] - \\
&\frac{1}{8} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{l}}^{i1}] - \frac{9}{16} m_1 g_1^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] + \frac{3}{16} m_1 g_1^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] - \\
&\frac{3}{8} m_1 g_1^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}] - \frac{9}{16} m_2 g_2^2 y_e^{pi2} y_e^{i1r} \overline{a_e}^{pr} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] - \frac{3}{16} m_2 g_2^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] + \\
&\frac{1}{4} \overline{y_e}^{st} y_e^{si2} y_e^{i1r} \overline{a_e}^{pr} a_e^{pt} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{4} \overline{y_e}^{st} y_e^{pi2} y_e^{i1t} \overline{a_e}^{pr} a_e^{sr} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] - \\
&\frac{1}{2} g_1^2 g_2^2 a_e^{i1i2} (m_1 + m_2) \mathrm{LF}_{1,1,1,1,1,-1}[m_1, m_2, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&\frac{1}{2} g_1^2 y_e^{pi2} \overline{a_e}^{pr} a_e^{i1r} \mathrm{LF}_{1,1,1,1,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}] + \\
&g_1^2 y_e^{i1r} \overline{a_e}^{pr} a_e^{pi2} \mathrm{LF}_{1,1,1,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] + \\
&m_1 g_1^2 \overline{a_e}^{pr} a_e^{pi2} a_e^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{2} g_2^2 y_e^{pi2} \overline{a_e}^{pr} a_e^{i1r} \mathrm{LF}_{1,1,1,1,1,-1}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}]\Big)
\end{aligned}
$$